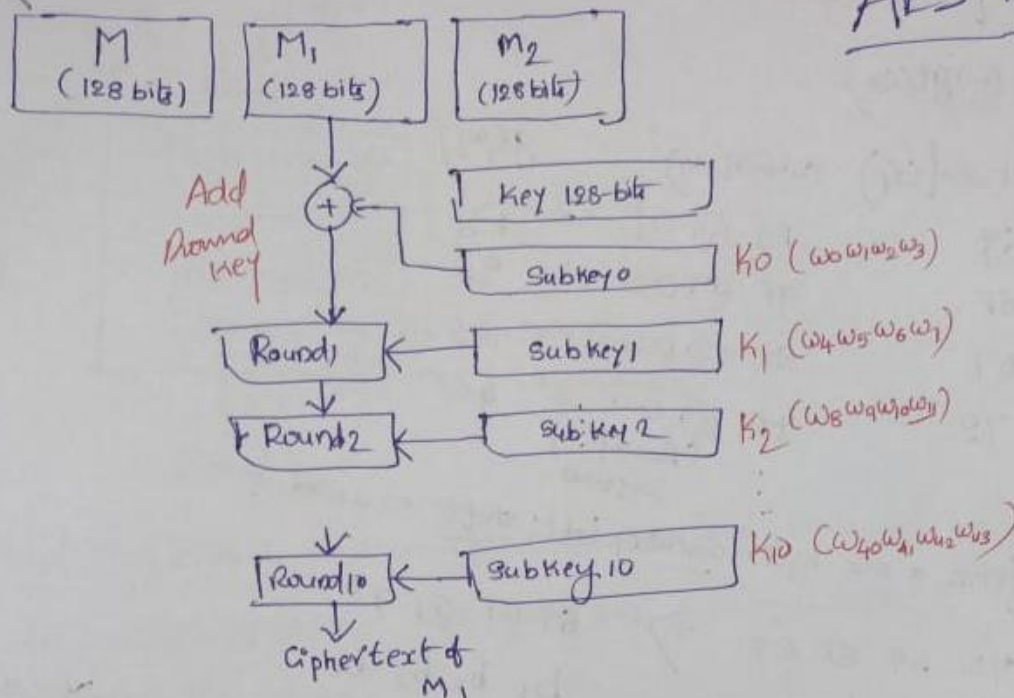
AES-128

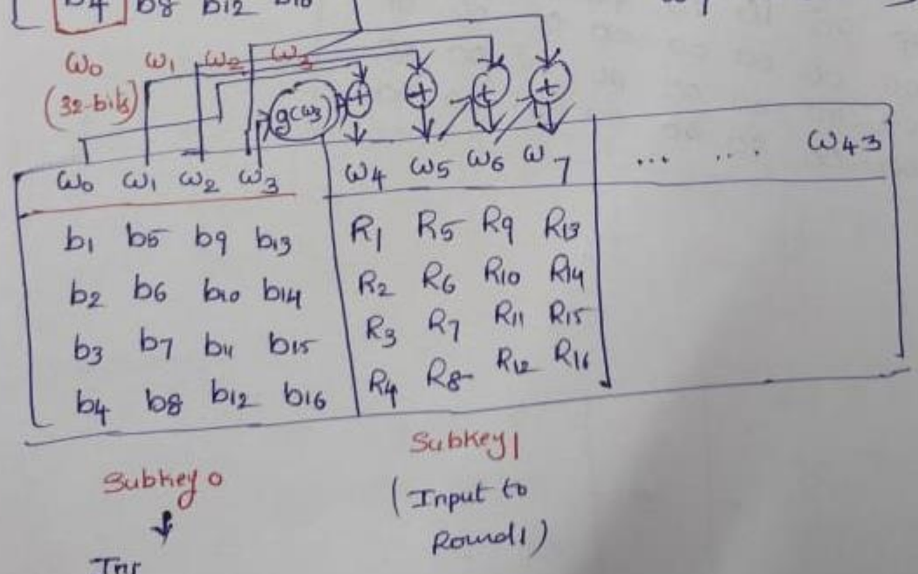


Key In Hex: 73 61 74 69 73 68 63 6A 69 73 62 6F 72 69 6E 67

b_1 b_2 b_3 b_4 b_5 b_6 b_7 b_8 b_9 b_{10} b_{11} b_{12} b_{13} b_{14} b_{15} b_{16}

$$\begin{bmatrix} b_1 & b_5 & b_9 & b_{13} \\ b_2 & b_6 & b_{10} & b_{14} \\ b_3 & b_7 & b_{11} & b_{15} \\ b_4 & b_8 & b_{12} & b_{16} \end{bmatrix} \Rightarrow$$

$$W_4 = W_0 \oplus g(\omega_3)$$
$$W_5 = \omega_4 \oplus \omega_1$$
$$\omega_6 = \omega_5 \oplus \omega_2$$
$$\omega_7 = \omega_6 \oplus \omega_3$$



What is this function g( ) ?

$W_4 = W_0 \oplus g(W_3)$

W_3	Rotword(X_1)	Subword(Y_1)		$g(W_3)$
72	69	F9 $\oplus$ 01	=	F8
69	6E	9F $\oplus$ 00	=	9F
6e	67	85 $\oplus$ 00	=	85
67	72	40 $\oplus$ 00	=	40

(Round constant)

→ Rotword performs a one-byte circular left shift on a word.

72 69 6E 67 ⇒ 69 6E 67 72

$b_0\ b_1\ b_2\ b_3$ ⇒ $b_1\ b_2\ b_3\ b_0$

→ Subword(Y_1) performs a byte substitution on each byte of its input word, using s-box. (Byte substitution)

→ The result of X_1 is Xored with a round constant, Rcon [j].

R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	R_{10}
01	02	04	08	10	20	40	80	1B	36
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00

Round 1

↓

Round 2

- Byte substitution
- Shift rows
- Mix columns
- Add round key ← Subkey$_2$ $(w_8\, w_9\, w_{10}\, w_{11})$

↓

Round 3

① **Byte substitution** does a simple replacement of each byte of the block data using an S-box. left four bits determine row, right four bits determine the column.

② **shift row**:- Simply byte shifts the rows.

- First row: no change
- Second row: one byte cyclical left shift
- Third row: two byte cyclical left shift.
- Fourth row: three byte cyclical left shift.

$$\begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \end{array}\begin{bmatrix} 63 & 47 & 63 & f0 \\ f2 & 9c & 7d & c5 \\ f0 & ab & 7b & 76 \\ af & 76 & 76 & ca \end{bmatrix} \xrightarrow{\text{No change}} \begin{bmatrix} 63 & 47 & 63 & F0 \\ 9c & 7d & c5 & F2 \\ 7b & 7C & F0 & ab \\ Ca & af & 76 & 76 \end{bmatrix}\begin{array}{c} \\ S_1 \\ S_2 \\ S_3 \end{array}$$

③ **Mixcolumns**:-

$$\underbrace{\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}}_{\text{Standard matrix}} \times \begin{bmatrix} 63 & 47 & 63 & F0 \\ 9c & 7d & C5 & F2 \\ 7b & 7C & F0 & ab \\ eA & AF & 76 & 76 \end{bmatrix} =$$

$$= \begin{bmatrix} r_1 & r_5 & r_9 & r_{13} \\ r_2 & r_6 & r_{10} & r_{14} \\ r_3 & r_7 & r_{11} & r_{15} \\ r_4 & r_8 & r_{12} & r_{16} \end{bmatrix} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} * \begin{bmatrix} 63 & 47 & 63 & F0 \\ 9C & 7d & C5 & F2 \\ 7b & 7C & F0 & AB \\ CA & AF & 76 & 76 \end{bmatrix}$$

$r_1 = 02 \times 63 + 03 \times 9C + 01 \times 76 + 01 \times CA$

$r_2 = 01 \times 63 + 02 \times 9C + 03 \times 7b + 01 \times CA$

$r_3 : \ldots$

$\vdots$

This is where the concept of finite field arithmetic comes into place.

$GF(2^8)$

$02 \Rightarrow 0000\ 0010$

$63 \Rightarrow 0110\ 0011$

$x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + 1$

The polynomial corresponding to 02 is $= X$ in $GF(2^8)$

" 63 is $= x^6 + x^5 + x + 1$

finite field multiplication

$02 \times 63 \Rightarrow x * (x^6 + x^5 + x + 1) \Rightarrow x^7 + x^6 + x^2 + 1 \Rightarrow \underbrace{1100}_{C}\underbrace{0110}_{6}$

$\boxed{02 \times 63 = C6}$

Note: The last round for encryption does not involve the "Mix columns" step

compute r_1 :— $02 * 63 + 03 * 9C + 01 * 7B + 01 * CA$

$03 * 9C \Rightarrow (\underbrace{0000}_{0}\ \underbrace{0011}_{3}) * (\underbrace{1001}_{9}\ \underbrace{1100}_{C})$

$\ldots x^7 \ldots x^4 + x^3 + x^2)$

(3)

$02 \times 63 \Rightarrow (0000\ 0010)(0110\ 0011) \Rightarrow x(x^6+x^5+x+1) +$

$03 \times 9C \Rightarrow (0000\ 0011)(1001\ 1100) \Rightarrow (x+1)(x^7+x^4+x^3+x^2) +$

$01 \times 7B \Rightarrow (0000\ 0001)(0111\ 1011) \Rightarrow 1 * (x^6+x^5+x^4+x^3+x+1) +$

$01 \times CA \Rightarrow (0000\ 0001)(1100\ 1010) \Rightarrow 1 * (x^7+x^6+x^3+x)$

$$\cancel{x^7}+\cancel{x^6}+\cancel{x^2}+\cancel{x}+x^8+\cancel{x^5}+\cancel{x^4}+\cancel{x^3}+x^7+x^4+\cancel{x^3}+\cancel{x^2}+\cancel{x^6}+\cancel{x^5}+\cancel{x^4}+\cancel{x^3}+x+1$$
$$+\cancel{x^7}+x^6+\cancel{x^3}+\cancel{x}$$

result polynomial :-

$$\Rightarrow \left(x^8+x^7+x^6+x^4+x+1\right) \quad GF(2^8)$$

all the polynomials within the range $[0, \ldots \cdot x^7+x^6+x^5+x^6+x^4+x^3+x^2+x+1]$

The resut polynomial is not in the range of $GF(2^8)$. Hence, it is divided by an Irreducible polynomial $(p(x))$.

$$\boxed{p(x) = x^8+x^4+x^3+x+1}$$

Note: In round10, there is no mix column step

$x^8+x^7+x^6+x^4+x+1 \Rightarrow 111010011$

$x^8+x^4+x^3+x+1 \Rightarrow 100011011$

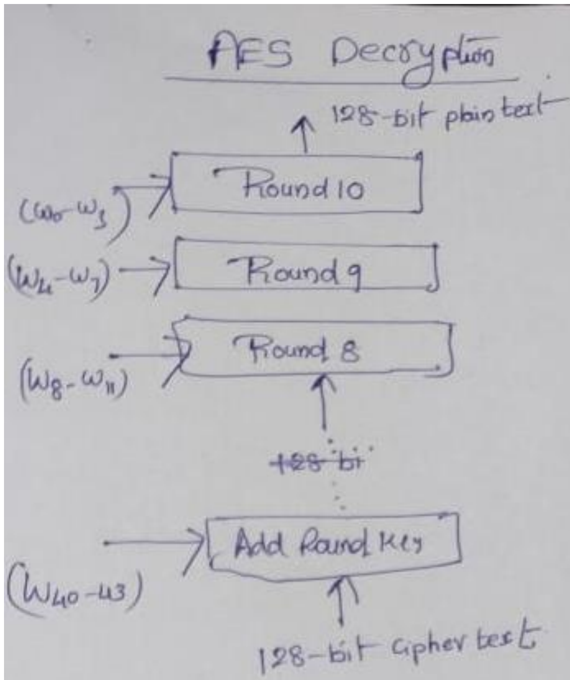
$$\begin{array}{r|l} 100011011 & 111010011 \\ & 100011011 \quad \oplus \\ \hline & 011001000 \end{array}$$

$\underline{1100}\ \underline{1000}$
C 8

$$\boxed{r_1 = C8}$$

After mix columns step, final step in the round function is add round key, which is xoring of respective key with the result to generate cipher text -

AES Decryption



Decryption - Round

Round has the following steps:

- Substitution Bytes
- Shift rows
- Mixing columns (not applicable for Round 10)
- Add round key.

- **Substitution bytes:-** we use Inverse S-box
- **Shift rows:-** Rows are shifted right in decryption

  1^{st} row – unchanged

  2^{nd} row shifted right by 1

  3^{rd} row shifted right by 2

  4^{th} row shifted right by 3

- **Mixcolumns:-** use another standard matrix

$$\begin{bmatrix} 0E & 0B & 0D & 09 \\ 09 & 0E & 0B & 0D \\ 0D & 09 & 0E & 0B \\ 0B & 0D & 09 & 0E \end{bmatrix} \times \begin{bmatrix} r_1 & \cdot & \cdot & \cdot \\ r_2 & \cdot & \cdot & \cdot \\ r_3 & \cdot & \cdot & \cdot \\ r_4 & \cdot & \cdot & r_{16} \end{bmatrix}$$

- **Add round key:** respective round key